

MILESTONE 3 & 4:

Milestone 3: Week 5 & 6 — Product Intelligence & Progress Tracking

This milestone integrates the **Ingredient Intelligence Module**, **Product Recommendation Engine**, **Progress Tracking System**, and **User/Consultant/Dermatologist Dashboards**.

Step 1: Implement the Ingredient Intelligence Module (**INGREDIENT INTELLIGENCE SERVICE**)

1. **Database Schema & Data Ingestion:**
 - Configure MongoDB document database collections for storing rich ingredient metadata (e.g., Retinoids, Niacinamide, Vitamin C, Hyaluronic Acid, Salicylic Acid, Ceramides, Peptides, AHAs/BHAs).
 - Ingest information including pH ranges, contraindications, synergies, and allergy tags.
2. **INCI Text Parsing Engine:**
 - Build an INCI parser using regex and NLP string processing to convert raw product ingredient strings into tokenized arrays.
3. **Interaction & Contraindication Analysis:**
 - Build endpoint **POST /api/v1/ingredients/analyze-compatibility** to evaluate active ingredient conflicts (e.g., flagging concurrent usage of Retinoids and AHAs/BHAs due to irritation risks).
4. **Allergy & Sensitivity Engine:**
 - Compare parsed product components directly against user profile sensitivity arrays (e.g., fragrance, specific chemical filters) and assign a safety rating (**Safe**, **Caution**, or **Unsafe**).
5. **Educational Context Generation:**
 - Provide response payloads explaining ingredient functions, benefits, and usage warnings.

Step 2: Implement the Product Recommendation Engine (**PRODUCT RECOMMENDATION SERVICE**)

1. Vector Search Setup (Vector DB):

- Generate 128-dimensional vector embeddings for user skin profiles and skincare products.
- Store and index vectors in a Vector DB (FAISS / Pinecone) for high-dimensional similarity matching.

2. Multi-Stage Scoring Pipeline:

- **Stage 1 (Similarity Search):** Compute cosine similarity between user vectors and product vectors to isolate the top candidate set.
- **Stage 2 (ML Ranking Model):** Pass top candidates through XGBoost / LightGBM models to compute refined product suitability scores.
- **Stage 3 (Hard Constraint Filtering):** Remove items flagged with user allergens or conflicting active ingredients.

3. Budget-Based Optimization:

- Build endpoint `POST /api/v1/products/recommend-routine-set` implementing a budget constraint solver across product categories (Face Wash, Serum, Moisturizer, Sunscreen, Toner, Face Masks)

$$\text{Maximize } \sum \text{SuitabilityScore}_i \quad \text{subject to} \quad \sum \text{Price}_i \leq \text{Budget}_{\text{max}}$$

4. Product Comparison & Alternative Generation:

- Create comparison endpoints allowing users to compare two products side-by-side or auto-generate lower-cost alternatives (duplicates).

Step 3: Implement Progress Tracking & Health Scoring (**HEALTH SCORING SERVICE & PROGRESS TRACKING SERVICE**)

1. Weighted Health Scoring Engine:

- Calculate the **Skin Health Score** dynamically using the weighted scoring model:
$$\{\text{Skin Health Score}\} = (0.35 \text{ times}\{\text{Condition Assessment}\}) + (0.20 \text{ times}\{\text{Lifestyle}\}) + (0.15 \text{ times}\{\text{Sleep}\}) + (0.20 \text{ times}\{\text{Routine Consistency}\}) + (0.10 \text{ times}\{\text{Hydration}\})$$

2. Progress Logging & Storage:

- Implement endpoint `POST /api/v1/progress/log-entry` to store daily completion flags, hydration metrics, and self-reported concerns.
- Upload progress photos securely to Cloud Storage (AWS S3 / Azure Blob) and link reference keys in the MongoDB `progress_logs` collection.

3. Analytics & Trend Analysis:

- Calculate 7-day, 30-day, and 90-day progress metrics mapping score variance against routine adherence levels.

Step 4: Build Interactive Dashboards (REACT.JS + TAILWIND CSS)

1. User Dashboard (`/dashboard/user`):

- **Score Overview:** Dynamic gauge showing current Skin Health Score and component breakdowns.
- **Skincare Checklist:** Daily interactive morning/evening routine checklists.
- **Product Recommendations:** Filterable cards listing top product matches with direct purchase/view actions.
- **Progress Charts:** Graphical visualization using `Chart.js` / `Plotly`.

2. Skincare Consultant Dashboard (`/dashboard/consultant`):

- Overview of assigned clients, assessment records, adherence ratings, and routine adjustment controls.

3. Dermatologist Dashboard (`/dashboard/dermatologist`):

- Detailed clinical views featuring high-resolution progress image comparisons, risk factor analyses, concern severity trends, and treatment note interfaces.

Milestone 4: Week 7 & 8 – Testing, Containerization, Deployment & Finalization

This milestone focuses on **notifications, administration, testing, security, Docker containerization, CI/CD, and production deployment.**

Step 1: Notification & Reminder System – NOTIFICATION SERVICE

1. **Background Jobs:**
 - Configure **Celery** + **Redis** for scheduled and background tasks.
2. **Notification Workflows:**
 - Routine reminders for morning/evening skincare.
 - Replenishment alerts when product volume falls below **10%**.
 - Daily hydration and sleep reminders.

Step 2: Admin Dashboard & Report System – ADMIN & REPORT SERVICE

1. **Admin Dashboard:**
 - User management and RBAC.
 - Active users and system health.
 - API performance and recommendation metrics.
2. **Report Generation:**
 - Implement **POST /api/v1/reports/generate**.
 - Generate **PDF/Excel** reports using ReportLab/Puppeteer and OpenPyXL/Pandas.
 - Include skin assessment, routines, products, and progress analytics.

Step 3: Testing, Security & Optimization:

1. **Testing:**
 - Unit testing with **pytest**.
 - API testing using **Postman/httpx**.
 - End-to-end flow:

2. Signup → Skin Assessment → Routine → Recommendation → Progress → Report
3. **Security:**
 - Implement **JWT/OAuth2 + RBAC**.
 - Secure database connections using **SSL/TLS**.
 - Protect sensitive files and cloud storage.
4. **Optimization:**
 - Use **Redis caching** for frequently accessed data.
 - Optimize API response time and database queries.
 -

Step 4: Docker Containerization

1. **Docker Configuration**
 - Create Dockerfiles for **FastAPI backend** and **React frontend**.
 - Use multi-stage builds where appropriate.
2. **Docker Compose – Recommended**
 - Configure required services such as PostgreSQL, Redis, MongoDB, and Vector DB.
 - Ensure proper service networking and environment configuration.
3. **Validation**
 - Build and run Docker containers successfully.
 - Verify API, frontend, database, cache, and other required services.
 - Confirm service health and communication.

So: Docker is **mandatory**, while Docker Compose can be **recommended/required only if your project has multiple services that need to run together**.

Step 5: CI/CD & Production Deployment:

1. **GitHub Actions:**
 - Trigger pipeline on push to **main**.
 - Run **flake8 + pytest**.
 - Build and tag Docker images.
 - Push images to a container registry.
2. **Cloud Deployment:**
 - Deploy using **AWS / Azure / Google Cloud / Render**.
 - Configure production database, storage, environment variables, and secrets.

3. **Monitoring:**

- Configure **Cloud Monitoring / CloudWatch / Grafana**.
- Monitor API health, latency, CPU/Memory, errors, and logs.

Step 6: Final Integration & Submission

1. Perform complete **end-to-end system testing**.
2. Fix critical bugs and verify all APIs.
3. Update **README, API documentation, setup instructions, and architecture diagram**.
4. Submit:
 - GitHub repository
 - Deployment URL
 - API documentation
 - Docker configuration
 - CI/CD workflow
 - Test results
 - Screen recording/demo of the complete system.

Outcomes for Milestone 3 (Week 5 & 6)

- **Product Recommendation Engine Operational:** The recommendation service is fully functional, capable of evaluating product suitability and filtering options based on user preferences and budget.
- **Progress Tracking System Functional:** The progress tracking workflows are active, allowing users to log skincare habits, track routine adherence, and monitor score changes over time.
- **Analytics and Recommendation Workflows Completed:** Analytics services are integrated to generate insights, trend reports, and customized product/ingredient recommendations.

Outcomes for Milestone 4 (Week 7 & 8)

- **Fully Deployed Production-Ready Platform:** The frontend and backend microservices are fully integrated, containerized with Docker, and deployed on cloud infrastructure (e.g., AWS or Azure).
- **Skincare Intelligence Systems Operational:** All core intelligence modules—including ingredient analysis, skin scoring, and recommendation models—are operating live in the production environment.

- **Complete End-to-End Skincare Planning Workflow Demonstrable:** The entire user journey (from skin assessment and routine generation to progress tracking and report exports) is fully functional and ready to be demonstrated across all user roles.